Introduction to Artificial Intelligence

Semester I, 2025-26

Modeling Uncertainty

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Outline

- Last Class
  - Informed Search
- This Class
  - Modeling Uncertainty
- Reference Material
  - AIMA Ch. 13 and 14

Acknowledgement

These slides are intended for teaching purposes only. Some material has been used/adapted from web sources and from slides by Doina Precup, Dorsa Sadigh, Percy Liang, Mausam, Dan Klein, Anca Dragan, Nicholas Roy and others.

Uncertainty is Pervasive in AI

- Models till now assumed that everything was certain
- Exact knowledge of the agent's domain
- Actions had precise effects - they acted as they were intended.
  - i.e., if you are in state S1 and you execute action A you arrive at state S2.

- In reality, we must have the ability to deal with uncertainty.
  - We might not know the status of a road when we choose to drive down it.
  - We don't know what the patient may be suffering from.

- We need ways to model uncertainty and incorporate uncertainty in our decision making.
  - We try to leave earlier than usual if it has rained.

Uncertainty in the real world

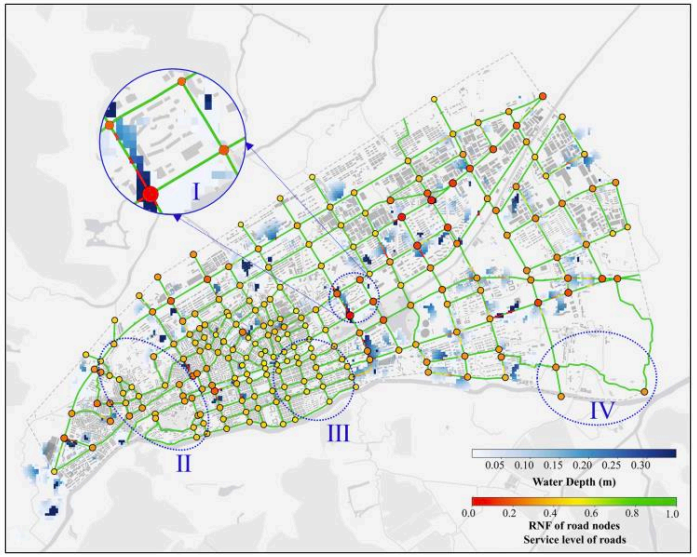
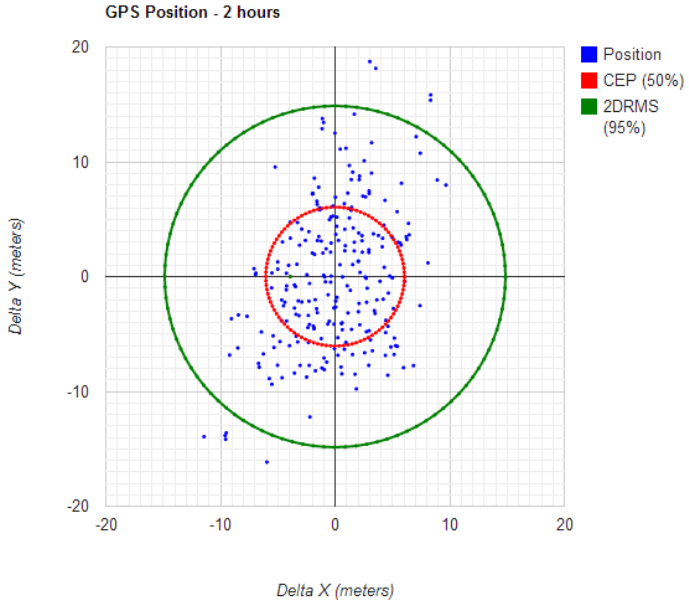
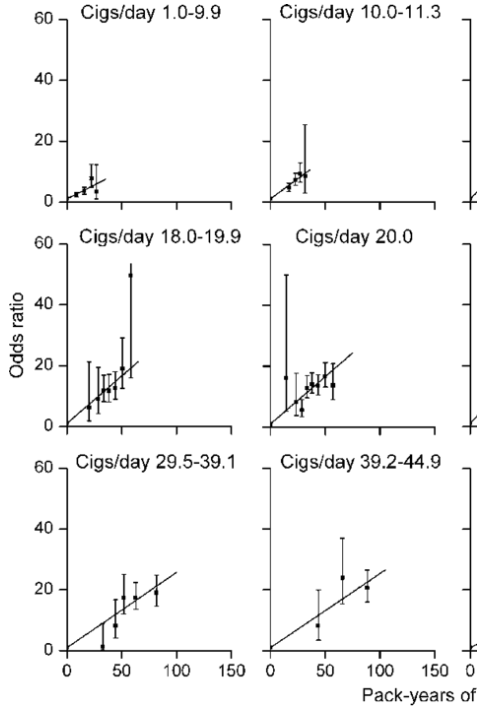
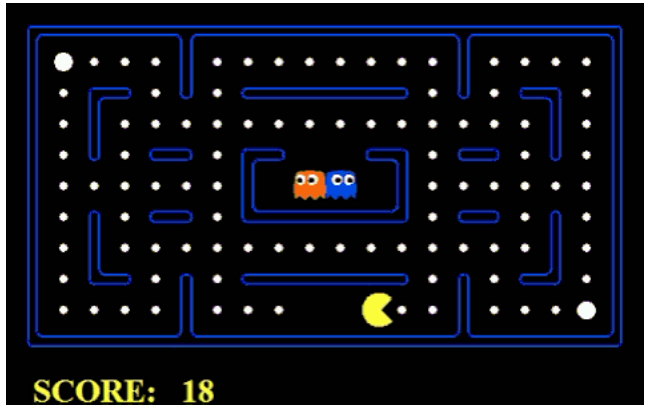


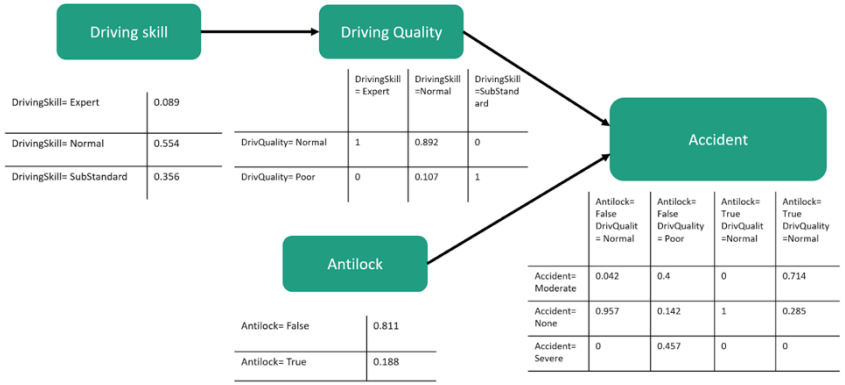
Fig 6. An inundation instance and concurrent RNF in Ling-Xi.











Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
  - R = Do I have Covid?
  - T = Engine is faulty or working?
  - D = How long will it take to drive to IIT?
  - L = Where is the person?

- Domains
  - R in {true, false} (often write as {+r, -r})
  - T in {faulty, working}
  - D in [0, ∞)
  - L in possible locations in a grid {(0,0), (0,1), …}

I hear an unusual sound and a burning smell in my car, what fault is there in my engine?

I have fever, loss of smell, loss of taste, do I have Covid?

I hear some footsteps in my house, where is the burglar?

Probability Distributions over Finite Sets

- A probability is a function defined over a set of atomic events U.
- U represents the universe of all possible events.
- Given U (a universe of events), a probability function is a function defined over subsets of U that maps each subset onto the real numbers
  - that satisfies the Axioms of Probability.

1. $P(U) = 1$

2. $P(A) \in [0,1]$

3. $P(\{\}) = 0$

4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

NB: if $A \cap B = \{\}$ then $P(A \cup B) = P(A) + P(B)$

Probability Distributions over Finite Sets

$A \vee B$: to represent the set of events with either property A or B, i.e. the set $A \cup B$

$A \wedge B$: to represent the set of events both property A and B, i.e. the set $A \cap B$

$\neg A$: to represent the set of events that do not have property A: the set U-A (i.e., the complement of A w.r.t. the universe of events U)

Probability Distributions over Finite Sets

We will model sets of events in our universe as vectors of feature values.

these are the constraint satisfaction problem

Like CSPs, we have

1. a set of variables $V_1, V_2, \ldots, V_n$

2. a finite domain of values for each variable, $Dom[V_1], Dom[V_2], \ldots, Dom[V_n]$.

The universe of events U is the set of all vectors of values for the variables

$$\langle d_1, d_2, \ldots, d_n \rangle : d_i \in Dom[V_i]$$

When we write P(A=a, B=b), we will mean the probability that variable A has been assigned value 'a' **and** variable B has been assigned value 'b'. Note that here, sets of events are induced by a given value assignment. So, P(A=a) represents a set of events in which A holds the value 'a'.

Probability Distributions over Finite Sets

Our event space has size $\prod_i |Dom[V_i]|$, i.e., the product of the domain sizes. If $|Dom[V_i]| = 2$, we have 2^n distinct atomic events.

Note the size of possible event outcomes (or variable assignments) grows **exponentially** with the number of variables.

Probability Distributions over Finite Sets

We often want to look at subsets of U defined by value assignments to particular variables.

E.g.

$\{V_1 = a\}$ is the set of all events where $V_1 = a$

$\{V_1 = a, V_3 = d\}$ is the set of all events where $V_1 = a$ and $V_3 = d$.

Probability Distributions over Finite Sets

If we have probability of every atomic event (wherein every event is a full instantiation of the variables) we can compute the probability of any other set of events.

E.g.

$\{V_1 = a\}$ is the set of all events where $V_1 = a$

$P(\{V_1 = a\}) =$

$$\sum_{x_2 \in Dom[V_2]} \sum_{x_3 \in Dom[V_3]} \sum_{x_4 \in Dom[V_4]} \ldots \sum_{x_n \in Dom[V_n]}$$

$P(\{V_1 = a, V_2 = x_2, V_3 = x_3, V_4 = x_4 \ldots, V_n = x_n\})$.

Probability Distributions over Finite Sets

Example:

$$P(\{V_1 = 1\}) = \sum_{x_2 \in Dom[V_2]} \sum_{x_3 \in Dom[V_3]} P(\{V_1 = 1, V_2 = x_2, V_3 = x_3\}).$$

(V1 = 1, V2 = 1, V3 = 1)	(V1 = 2, V2 = 1, V3 = 1)	(V1 = 3, V2 = 1, V3 = 1)
(V1 = 1, V2 = 1, V3 = 2)	(V1 = 2, V2 = 1, V3 = 2)	(V1 = 3, V2 = 1, V3 = 2)
(V1 = 1, V2 = 1, V3 = 3)	(V1 = 2, V2 = 1, V3 = 3)	(V1 = 3, V2 = 1, V3 = 3)
(V1 = 1, V2 = 2, V3 = 1)	(V1 = 2, V2 = 2, V3 = 1)	(V1 = 3, V2 = 2, V3 = 1)
(V1 = 1, V2 = 2, V3 = 2)	(V1 = 2, V2 = 2, V3 = 2)	(V1 = 3, V2 = 2, V3 = 2)
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(V1 = 1, V2 = 3, V3 = 1)	(V1 = 2, V2 = 3, V3 = 1)	(V1 = 3, V2 = 3, V3 = 1)
(V1 = 1, V2 = 3, V3 = 2)	(V1 = 2, V2 = 3, V3 = 2)	(V1 = 3, V2 = 3, V3 = 2)
(V1 = 1, V2 = 3, V3 = 3)	(V1 = 2, V2 = 3, V3 = 3)	(V1 = 3, V2 = 3, V3 = 3)

Probability Distributions over Finite Sets

Example:

$$P(\{V_1 = 1, V_3 = 2\}) = \sum_{x_2 \in Dom[V_2]} P(\{V_1 = 1, V_2 = x_2, V_3 = 2\}).$$

(V1 = 1, V2 = 1, V3 = 1)	(V1 = 2, V2 = 1, V3 = 1)	(V1 = 3, V2 = 1, V3 = 1)
(V1 = 1, V2 = 1, V3 = 2)	(V1 = 2, V2 = 1, V3 = 2)	(V1 = 3, V2 = 1, V3 = 2)
(V1 = 1, V2 = 1, V3 = 3)	(V1 = 2, V2 = 1, V3 = 3)	(V1 = 3, V2 = 1, V3 = 3)
(V1 = 1, V2 = 2, V3 = 1)	(V1 = 2, V2 = 2, V3 = 1)	(V1 = 3, V2 = 2, V3 = 1)
(V1 = 1, V2 = 2, V3 = 2)	(V1 = 2, V2 = 2, V3 = 2)	(V1 = 3, V2 = 2, V3 = 2)
(V1 = 1, V2 = 2, V3 = 3)	(V1 = 2, V2 = 2, V3 = 3)	(V1 = 3, V2 = 2, V3 = 3)
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(V1 = 1, V2 = 3, V3 = 2)	(V1 = 2, V2 = 3, V3 = 2)	(V1 = 3, V2 = 3, V3 = 2)
(V1 = 1, V2 = 3, V3 = 3)	(V1 = 2, V2 = 3, V3 = 3)	(V1 = 3, V2 = 3, V3 = 3)

Joint Distributions

- A **joint distribution** over a set of random variables: $X_1, X_2, \ldots X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$

$$P(x_1, x_2, \ldots x_n)$$

  - Must obey:

$$P(x_1, x_2, \ldots x_n) \geq 0$$

$$\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Note: Joint distribution can answer all probabilistic queries.

Problem: Table size is d^n.

Events

- An event is a set E of outcomes

$$P(E) = \sum_{(x_1 \ldots x_n) \in E} P(x_1 \ldots x_n)$$

- From a joint distribution, we can calculate the probability of any event
  - Probability that it's hot AND sunny? .4
  - Probability that it's hot? .4 + .1
  - Probability that it's hot OR sunny? .4 + .1 + .2

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

General form

$$\Pr(X_i) = \sum_{X_1} \dots \sum_{X_{i-1}} \sum_{X_{i+1}} \dots \sum_{X_n} \Pr(X_1, X_2, \dots, X_n)$$

Marginalization

- From a joint distribution (>1 variable) reduce it to a distribution over a smaller set of variables
- Called marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding likelihoods

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(t) = \sum_s P(t, s)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$$P(s) = \sum_t P(t, s)$$

$P(W)$

W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

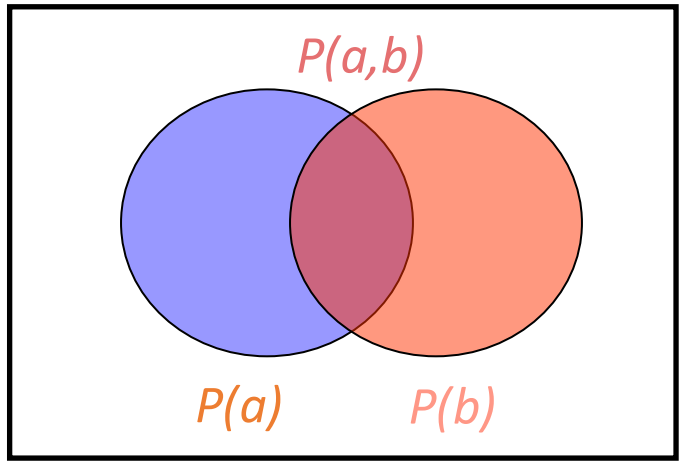
General form

$$Pr(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n | X_i) = \frac{P(X_1,\dots,X_n)}{P(X_i)} = \frac{P(x_1,\dots,X_n)}{\sum_{X_1} \dots \sum_{X_{i-1}} \sum_{X_{i+1}} \dots \sum_{X_n} \Pr(X_1,\dots,X_n)}$$

Conditioning

- Conditional distributions are probability distributions over some variables given fixed values of others

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$P(T, W)$ Joint Distribution

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

$P(W|T)$

$P(W|T = cold)$

W	P
sun	0.4
rain	0.6

$P(W|T = hot)$

W	P
sun	0.8
rain	0.2

Inference by Enumeration

- P(W)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- P(W)?

P(sun)=.3+.1+.1+.15=.65

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- P(W)?

P(sun)=.3+.1+.1+.15=.65
P(rain)=1-.65=.35

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- P(W | winter, hot)?

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Inference by Enumeration

- P(W | winter, hot)?

P(sun|winter,hot)~.1
P(rain|winter,hot)~.05
P(sun|winter,hot)=2/3
P(rain|winter,hot)=1/3

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Chain Rule

Chain rule is derived by successive application of product rule:

$$
\begin{aligned}
& P(X_1, \ldots, X_n) = \\
= \quad & P(X_1, \ldots, X_{n-1}) P(X_n | X_1, \ldots, X_{n-1}) \\
= \quad & P(X_1, \ldots, X_{n-2}) P(X_{n-1} | X_1, \ldots, X_{n-2}) P(X_n | X_1, \ldots, X_{n-1}) \\
= \quad & \ldots \\
= \quad & \prod_{i=1}^{n} P(X_i | X_1, \ldots, X_{i-1})
\end{aligned}
$$

Constructing a larger distribution by simpler distribution.

Bayes Rule



- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)}{P(y)}P(x)$$

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Usefulness
  - Lets us build one conditional from its reverse.
  - Often one conditional is difficult to obtain but the other one is simple.

Independence

- Two variables are *independent* if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

  - This says that their joint distribution *factors* into a product two simpler distributions
  - Another form: $\forall x, y : P(x|y) = P(x)$
  - We write: $X \perp\!\!\!\perp Y$
  - Examples
    - N-independent flips of a fair coin.
    - P(*bus-arrives-on-time* | *wearing-blue-shirt)* = P(*bus-arrives-on-time*)

n smaller distributions

$P(X_1)$

H	0.5
T	0.5

$P(X_2)$

H	0.5
T	0.5

$\vdots$

$P(X_n)$

H	0.5
T	0.5

$P(X_1, X_2, \ldots X_n)$

2^n

Conditional Independence

- X and Y are conditionally independent given Z.
- Given Z , X does not add information about Y and vice versa.

$$P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

$$P(x \mid z) = P(x \mid z, y)$$

$$P(y \mid z) = P(y \mid z, x)$$

Example:

- Variable A: Student 1 gets an A on the exam.
- Variable B: Student 2 gets an A on the exam.
- Variable C (Condition): The official difficulty level of the exam (e.g., Easy vs. Hard)

Samples from a Distribution

- Sampling from given distribution
  - Step 1: Get sample *u* from uniform distribution over [0, 1)
  - Step 2: Convert this sample *u* into an outcome for the given distribution by having each outcome associated with a sub-interval of [0,1) with sub-interval size equal to probability of the outcome
- Utility? We should be able to sample from a CPT defining the probabilistic model.
- Next, we look at approaches for sampling from a Bayes Net.

Example:

C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = red$$
$$0.6 \leq u < 0.7, \rightarrow C = green$$
$$0.7 \leq u < 1, \rightarrow C = blue$$

- If random() returns *u* = 0.83, then our sample is *C* = blue
- E.g, after sampling 8 times:

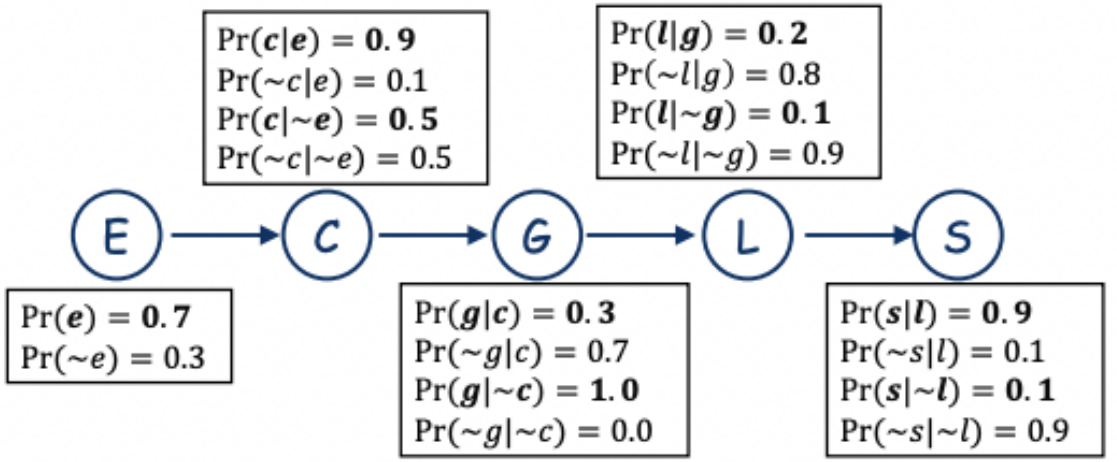
Example: Sequence of probabilistic relationships

If Pascal woke up too early E, Pascal probably needs coffee C; if Pascal needs coffee, he's likely grumpy G. If he is grumpy then it's possible that the lecture won't go smoothly L. If the lecture does not go smoothly then the students will likely be sad S.



E - Pascal woke up too early G - Pascal is grumpy S - Students are sad
C - Pascal needs coffee L- The lecture did not go smoothly

-



Sometimes we get samples by simply taking actions

Follow a route and see how much time it takes to reach the goal.

Randomly play a game of chess.

Explore the frontier randomly while searching.

Sampling interactions (as we will see later) will lead to a certain form of "learning"

https://www.youtube.com/watch?v=TmPfTpjtdgg

Using samples to build estimates

Goal: Compute expected age of students enrolled in this class

Known P(A)

$$E[A] = \sum_a P(a) \cdot a \qquad = 0.35 \times 20 + \ldots$$

Without P(A), instead collect samples [a_1, a_2, … a_N]

Unknown P(A): "Model Based" way

$$\hat{P}(a) = \frac{\text{num}(a)}{N}$$

$$E[A] \approx \sum_a \hat{P}(a) \cdot a$$

Eventually, we learn the right model which gives the right estimates.

Unknown P(A): "Model Free" way

$$E[A] \approx \frac{1}{N} \sum_i a_i$$

Bypass the model construction. Averaging works because the samples appear with the right frequencies.

Takeaways

- Uncertainty is inherent in the real world.
- We want to (i) model uncertainty and (ii) incorporate the uncertainty in our decision making.
- This lecture is about modeling uncertainty – use of probabilities and random variables. Gives us tools to perform estimation.
- Important to think of the distribution over variables as a way to represent how entities we want to reason over are interacting with each other.
- Sometime we may not know about the distribution, we may only have samples from it. From samples we may need to estimate what underlying distribution the data emerged from.